

Solutions to MP363 Problem Set 1

(1)

P.1

The point of this problem is to show that even though the Rayleigh-Jeans spectral radiance formula

$$B_{RJ}(\nu, T) = \frac{2\nu^2}{c^2} k_B T$$

isn't fully correct for all values of ν , it does work pretty well if ν is much smaller than $k_B T/h$. This is because, for ν in this range, the Planck formula

$$B(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}$$

is approximately equal to $B_{RJ}(\nu, T)$.

We see this by expanding $B(\nu, T)$ in a Taylor series around $\nu=0$.

Now, we could do this using the full defn of a Taylor series, i.e.

$$B(\nu, T) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{\partial^n B}{\partial \nu^n} \right|_{\nu=0} \nu^n.$$

However, it'll turn out that $B(0, T)$ and $\frac{\partial B}{\partial \nu}(0, T)$ are both zero, so that we'll need to compute $\frac{\partial^2 B}{\partial \nu^2}(0, T)$ and $\frac{\partial^3 B}{\partial \nu^3}(0, T)$ to find the first two non-zero terms of $B(\nu, T)$. However, there's an easier way to do this: if we find the Taylor series of $f(\nu, T) = \frac{\nu}{e^{h\nu/k_B T} - 1}$, and then multiply by $2h\nu^2/c^2$, we'll get the Taylor series for $B(\nu, T)$.

(much like if we wanted to Taylor-expand $\frac{x^2}{1-x}$, we could use

$$\frac{1}{1-x} = 1+x+x^2+\dots \text{ to get } \frac{x^2}{1-x} = x^2(1+x+x^2+\dots) \\ = x^2+x^3+x^4+\dots)$$

So we need $f(0, T)$ and $\frac{\partial f}{\partial \nu}(0, T)$ to get the first two terms

$$f(\nu, T) = f(0, T) + \frac{\partial f}{\partial \nu}(0, T) \nu + O(\nu^2).$$

We'll need l'Hôpital's rule, which says that if we want to find

$\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$ when both p and q vanish at $x=a$, the limit is

the same as $\lim_{x \rightarrow a} \frac{p'(x)}{q'(x)}$. If either $p'(x)$ or $q'(x)$ is non-zero at $x=a$, we have the limit. If they both vanish, do it again.

$f(0, T)$ is such a limit:

$$\lim_{\nu \rightarrow 0} f(\nu, T) = f(0, T) = \lim_{\nu \rightarrow 0} \frac{\nu}{e^{h\nu/k_B T} - 1} = \lim_{\nu \rightarrow 0} \frac{\frac{d}{d\nu} \nu}{\frac{d}{d\nu} (e^{h\nu/k_B T} - 1)}$$

$$= \lim_{\nu \rightarrow 0} \frac{1}{\frac{h}{k_B T} e^{h\nu/k_B T}} = \frac{k_B T}{h}$$

so $f(0, T) = k_B T/h$. Now, to find $\frac{\partial f}{\partial \nu}(0, T)$, we obviously need $\frac{\partial f}{\partial \nu}(\nu, T)$, which is easily computed using the quotient rule:

$$\frac{\partial f}{\partial \nu}(\nu, T) = \frac{\partial}{\partial \nu} \left[\frac{\nu}{e^{h\nu/k_B T} - 1} \right] = \frac{(\frac{\partial}{\partial \nu} \nu)(e^{h\nu/k_B T} - 1) - \nu \frac{\partial}{\partial \nu} (e^{h\nu/k_B T} - 1)}{(e^{h\nu/k_B T} - 1)^2}$$

$$= \frac{e^{h\nu/k_B T} - 1 - \frac{h\nu}{k_B T} e^{h\nu/k_B T}}{(e^{h\nu/k_B T} - 1)^2}$$

As $\nu \rightarrow 0$, we see both top and bottom go to zero, so we need L'Hôpital's rule:

$$\frac{\partial f}{\partial \nu}(\nu, T) = \lim_{\nu \rightarrow 0} \frac{e^{h\nu/k_B T} - 1 - \frac{h\nu}{k_B T} e^{h\nu/k_B T}}{(e^{h\nu/k_B T} - 1)^2} = \lim_{\nu \rightarrow 0} \frac{\frac{\partial}{\partial \nu} [e^{h\nu/k_B T} - 1 - \frac{h\nu}{k_B T} e^{h\nu/k_B T}]}{\frac{\partial}{\partial \nu} (e^{h\nu/k_B T} - 1)^2}$$

$$= \lim_{\nu \rightarrow 0} \frac{\frac{h}{k_B T} e^{h\nu/k_B T} - \frac{h}{k_B T} e^{h\nu/k_B T} - \frac{h^2 \nu}{k_B^2 T^2} e^{h\nu/k_B T}}{2h\nu/k_B T e^{h\nu/k_B T} (e^{h\nu/k_B T} - 1)}$$

$$= -\frac{h}{2k_B T} \lim_{\nu \rightarrow 0} \frac{\nu}{e^{h\nu/k_B T} - 1}$$

But we already showed that this limit was $\frac{k_B T}{h}$, so

$$\frac{\partial f}{\partial \nu}(\nu, T) = -\frac{h}{2k_B T} \left(\frac{k_B T}{h} \right) = -1/2$$

Thus, the Taylor series expansion for $\frac{\nu}{e^{h\nu/k_B T} - 1}$ around $\nu = 0$ is

$$\frac{\nu}{e^{h\nu/k_B T} - 1} = \frac{k_B T}{h} - \frac{1}{2} \frac{h\nu^2}{2k_B T} + O(\nu^2)$$

Thus, multiplying through by $\frac{2h\nu^2}{c^2}$ gives

$$B(\nu, T) = \frac{2\nu^2}{c^2} k_B T - \frac{h\nu^3}{c^2} + O(\nu^4)$$

as desired. Now, note that the ratio of the first and second terms are

$$\left(-\frac{h\nu^3}{c^2} \right) / \left(\frac{2\nu^2}{c^2} k_B T \right) = -\frac{h\nu}{2k_B T}$$

so if $\nu \ll k_B T/h$, the second term is much, much smaller than the first term (and all other terms of order 4 or more are even smaller still), so, if $\nu \ll k_B T/h$, then

$$B(\nu, T) \approx \frac{2\nu^2}{c^2} k_B T = B_{RJ}(\nu, T)$$

and so the Planck and RJ formulas agree for small ν . But how small is this ν ? Well, if we're looking at a black body at room temperature, say $T = 300 \text{ K}$, then

$$k_B T/h \approx (1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1})(300 \text{ K}) / (6.626 \times 10^{-34} \text{ J})$$

$$= 6.25 \times 10^{12} \text{ Hz},$$

which means the RJ formula is good enough for frequencies typical for frequencies below infrared - microwaves and radio waves. For a typical star, $T \sim 5000 \text{ K}$, so $k_B T/h = 1.04 \times 10^{14} \text{ Hz}$, which is in the visible region, so the RJ formula works fairly well for visible black bodies. However, $T = 2.7 \text{ K}$ is the temperature of the cosmic background radiation, which gives

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$k_B T/h = 5.62 \times 10^{10} \text{ Hz}$; this is in the microwave region, so the RJ formula will only be for very low frequencies, mostly radio waves. Luckily, the Planck formula is known and valid for all frequencies, so we can use it everywhere.

P.2

$B(\nu, T) d\nu$ gives the power output of a blackbody between frequencies ν and $\nu + d\nu$, if the body is at temperature T . However, we know that we can just as well use the wavelength λ to characterise the radiation, so there must be some function $f(\lambda, T)$ which tells us how much power flux there is between λ and $\lambda + d\lambda$, i.e. $f(\lambda, T) d\lambda$. Now, the amount of energy is the same no matter which we choose: whether we use ν or λ is our choice - the blackbody doesn't care, and puts out the same power either way. Thus, $|B(\nu, T) d\nu| = |f(\lambda, T) d\lambda|$. We use absolute values because $d\nu > 0 \Rightarrow d\lambda < 0$: if frequency goes up, wavelength goes down. Since we let $f(\lambda, T)$ take a positive quantity, we say $B(\nu, T) d\nu = -f(\lambda, T) d\lambda$. In other words, the extra power we detect by increasing the frequency from ν to $\nu + d\nu$ is the same as if we decreased the wavelength from λ to $\lambda - d\lambda$.

(a) To get $f(\lambda, T)$, we need to express everything in terms of λ : we know $\nu = c/\lambda$, so $B(\nu, T) = B(\frac{c}{\lambda}, T) = \frac{2hc}{\lambda^3} \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1}$. But $d\nu = \frac{d\nu}{d\lambda} d\lambda$ is how a change in λ gives a change in ν , and since $\nu = c/\lambda$, $\frac{d\nu}{d\lambda} = -\frac{c}{\lambda^2}$. Thus,

$$f(\lambda, T) d\lambda = -B(\frac{c}{\lambda}, T) \frac{d\nu}{d\lambda} d\lambda$$

$$= -\frac{2hc}{\lambda^3} \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1} \left(-\frac{c}{\lambda^2} d\lambda \right)$$

and so

$$f(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1}$$



is the spectral radiance per unit wavelength. Due to the $1/\lambda^5$, it's very sharply peaked about some maximum value λ_w , as shown.

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(b) Now try to find λ_w or at least some expression that defines it.

If it's the maximum $\frac{\partial f}{\partial \lambda}$ not vanishes $\lambda = \lambda_w$. The derivative is

$$\frac{\partial f(\lambda, T)}{\partial \lambda} = \frac{\partial}{\partial \lambda} \left[\frac{2hc^2}{\lambda^5} \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1} \right] = 2hc^2 \frac{\frac{\partial}{\partial \lambda} \left(\frac{1}{\lambda^5} \right) \left[\exp(\frac{hc}{\lambda k_B T}) - 1 \right] - \frac{1}{\lambda^5} \frac{\partial}{\partial \lambda} \left[\exp(\frac{hc}{\lambda k_B T}) - 1 \right]}{\left[\exp(\frac{hc}{\lambda k_B T}) - 1 \right]^2}$$

$$= 2hc^2 \frac{(-5\lambda^{-6}) \left(\exp(\frac{hc}{\lambda k_B T}) - 1 \right) - \frac{1}{\lambda^5} \left[-\frac{hc}{\lambda^2 k_B T} \exp(\frac{hc}{\lambda k_B T}) \right]}{\left[\exp(\frac{hc}{\lambda k_B T}) - 1 \right]^2}$$

$$= \frac{-2hc^2}{\lambda^7} \frac{5\lambda \exp(\frac{hc}{\lambda k_B T}) - 5 + \frac{hc}{k_B T} \exp(\frac{hc}{\lambda k_B T})}{\left[\exp(\frac{hc}{\lambda k_B T}) - 1 \right]^2}$$

At $\lambda = \lambda_w$, this not vanishes, which means the numerator is zero:

$$5\lambda_w \exp(\frac{hc}{\lambda_w k_B T}) - 5\lambda_w + \frac{hc}{k_B T} \exp(\frac{hc}{\lambda_w k_B T})$$

$$= 5\lambda_w \left(e^{\frac{hc}{\lambda_w k_B T}} - 1 \right) + \frac{hc}{k_B T} e^{\frac{hc}{\lambda_w k_B T}} = 0.$$

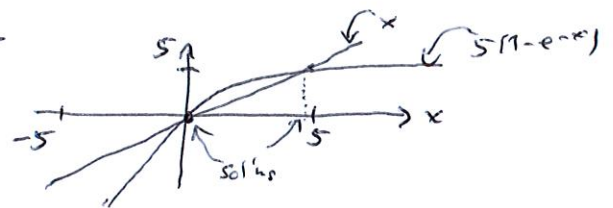
Now, for ease of notation, let $x = \frac{hc}{\lambda_w k_B T}$; then the above can be rewritten as

$$\frac{5hc}{x k_B T} (e^x - 1) + \frac{hc}{k_B T} e^x = 0.$$

Now, divide by $\frac{hc}{k_B T}$ and multiply through by x to get $5(e^x - 1) + xe^x = 0$, or $xe^x = 5(1 - e^{-x})$. Finally, multiply by e^{-x} to get

$$\boxed{x = 5(1 - e^{-x})}$$

as derived. To get a rough idea of what values of x satisfy this, plot x and $5(1 - e^{-x})$ on the same set of axes; where the curves cross gives the solution to the above eq'n.



(c) If we do this, we see that there

are two crossing points, at $x=0$ and at a value slightly below 5. Numerics

give a more precise result of $x \approx 4.965$. From the plot of $f(x, T)$,

you see that the maximum must occur at $x \approx 4.965$, and so

$$\lambda_w \text{ is defined by } x = \frac{hc}{\lambda_w k_B T} \approx 4.965.$$

$$\text{Thus, } \lambda_w T \approx \frac{hc}{4.965 k_B} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s}) (3 \times 10^8 \text{ m}\cdot\text{s}^{-1})}{4.965 (1.38 \times 10^{-23} \text{ J}\cdot\text{K}^{-1})} = \boxed{2.9 \times 10^{-3} \text{ m}\cdot\text{K}}$$

as expected. This is Wien's Displacement Law.

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(d) Now to actually use it: a typical star might have a surface temperature of $T = 5500\text{K}$, so if we apply λ_w for this, we get $\lambda_w = \frac{2.9 \times 10^{-3} \text{ m}\cdot\text{K}}{5500\text{K}} = 5.273 \times 10^{-7} \text{ m} = 527.3 \text{ nm}$. If you look up the wavelength-colour correspondence (I used the converter at academo.org/demos/wavelength-to-colour-relationship), you find that the star is yellowish-green.

(In reality, this does not mean that star is completely green - just because the maximum of $f(\lambda, T)$ is at this wavelength, all other frequencies are emitted as well, including those near this peak. As a result, the actual color is kind of an average of all these. To be precise, it would be

$$\lambda_{\text{avg}} = \frac{\int_0^\infty \lambda f(\lambda, T) d\lambda}{\int_0^\infty f(\lambda, T) d\lambda} = \frac{5.33 \times 10^{-3} \text{ m}\cdot\text{K}}{T}$$

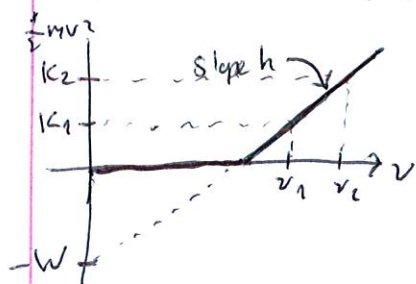
or about 970 nm , in the near-infrared part of the spectrum. So even though we can see the visible portion of its radiation, a "typical" wavelength would be in the infrared region.

P.3

The key to this is that the KE vs. frequency curve is determined by the energy conservation eqn given, i.e. $\frac{1}{2}mv^2 = h\nu - W$. Some see that the slope of the curve - a line, actually, is Planck's constant and the y-intercept - where it would cross the KE axis - is the negative of W . I say "would" because, since KE cannot be negative, the plot should actually be given by

$$\frac{1}{2}mv^2 = \begin{cases} 0 & h\nu - W < 0 \text{ (no photoemission)} \\ h\nu - W & h\nu - W \geq 0 \text{ (photoemission)} \end{cases}$$

i.e. like the below. However, as long as photoemission occurs,



we know it's a line, so all we need are two points. So if we know the electrons have KE K_1 at frequency ν_1 and K_2 at ν_2 , then

$$K_1 = h\nu_1 - W, \quad K_2 = h\nu_2 - W.$$

Two eqns for two unknowns, which can be solved:

$$h = \frac{K_2 - K_1}{\nu_2 - \nu_1}, \quad W = \frac{K_2\nu_1 - K_1\nu_2}{\nu_2 - \nu_1}$$

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so if we have two points, we set h and W .

(a) And we're given two points for gold photoemission: $\nu_1 = 2 \times 10^{15} \text{ Hz}$, $K_1 = 3.0 \text{ eV}$ and $\nu_2 = 10^{16} \text{ Hz}$, $K_2 = 36.1 \text{ eV}$. Thus, the gold work function W_{Au} is

$$W_{\text{Au}} = \frac{(36.1 \text{ eV})(2 \times 10^{15} \text{ Hz}) - (3.0 \text{ eV})(10^{16} \text{ Hz})}{10^{16} \text{ Hz} - 2 \times 10^{15} \text{ Hz}} = \boxed{5.28 \text{ eV}}$$

Planck's constant is

$$h = \frac{36.1 \text{ eV} - 3.0 \text{ eV}}{10^{16} \text{ Hz} - 2 \times 10^{15} \text{ Hz}} = 4.138 \times 10^{-15} \text{ eV} \cdot \text{s}$$

but we want it in J·s, so we use $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ to get

$$h = 4.138 \times 10^{-15} (1.6 \times 10^{-19} \text{ J}) \cdot \text{s} = \boxed{6.62 \times 10^{-34} \text{ J} \cdot \text{s}}$$

which is, of course, quite close to the real value.

(b) As we said, photoemission does not occur when $h\nu - W < 0$;

at which point the electron has zero kinetic energy (and thus the electrons can escape from the sample). $h\nu - W \geq 0$ implies $\nu \geq W/h$, i.e. $\nu \geq \nu_T$ where ν_T is the threshold frequency, the minimum frequency for photoemission, viz. W/h .

$W_K = 2.3 \text{ eV}$ for potassium, and using $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ above

gives

$$\nu_T = \frac{2.3 \text{ eV}}{6.62 \times 10^{-34} \text{ J} \cdot \text{s}} = \frac{2.3 \times 1.6 \times 10^{-19} \text{ J}}{6.62 \times 10^{-34} \text{ J} \cdot \text{s}} = \boxed{5.56 \times 10^{14} \text{ Hz}}$$

So only at 556 THz and above will we see photoelectric emission for potassium.

P. 4

The key to this problem lies in comparing the reduced mass of the given "atom" to that of a hydrogen atom. The Bohr model predicts

$$E_n^{(H)} = -\frac{\mu_H e^4}{8\epsilon_0^2 h^2} \frac{1}{n^2}$$

for the hydrogen atom, where $\mu_H = m_e m_p / (m_e + m_p)$ is the reduced electron-proton mass.

But Bohr's argument only works for any two-body system: if m_1 and

m_2 are the two particles, then the energy levels of the bound system must follow $E_n = -\frac{\mu e^4}{8\epsilon_0^2 h^2} \frac{1}{n^2}$, where

$\mu = \frac{m_1 m_2}{m_1 + m_2}$. Thus, if we compare the n^{th} level of this new "atom" to that of hydrogen, we find a

ratio of $E_n / E_n^{(H)} = \mu / \mu_H$. So the relative energies

⑦

depend on the reduced mass, not on the physical constants e , h , ϵ_0 or the level n .

(a) So for muonium, $m_1 = m_\mu$ and $m_2 = m_e$. Since $m_\mu = 207 m_e$, we get a reduced mass of

$$\mu^{(M)} = \frac{207 m_e \cdot m_e}{207 m_e + m_e} = \frac{207}{208} m_e \approx m_e$$

So $207/208$ is very close to 1 (0.995, to be precise). But the

hydrogen reduced mass is $\mu^{(H)} = \frac{1836 m_e \cdot m_e}{1836 m_e + m_e} = \frac{1836}{1837} m_e \approx m_e$.

Because the muon and proton are much more massive than the electron, $\mu^{(M)} \approx \mu^{(H)}$, and so $E_n^{(M)} / E_n^{(H)} \approx 1$, i.e.

The muonium and hydrogen energy levels are almost identical.

Ground state energy of muonium is -13.6 eV , the first excited state is -3.4 eV , then -1.51 eV , etc. Its spectral lines would be extremely close to those of hydrogen.

(b) The electron and positron have identical masses, so their

reduced mass is $\mu^{(P)} = \frac{m_e \cdot m_e}{m_e + m_e} = \frac{1}{2} m_e$. Since $\mu^{(H)} \approx m_e$, this gives $E_n^{(P)} / E_n^{(H)} \approx 1/2$, i.e. the energy levels of positronium are about half those of hydrogen.

The first level would have $-13.6/2 \text{ eV} = -6.8 \text{ eV}$, the next -1.7 eV , etc. All the spectral lines would be shifted down by a factor of two.